

Chemistry V2 — ExamSIDE Solution & Transfer Book

Attempt each source-faithful transfer item before opening support. H1 Notice → H2 Rule/Model/Representation → H3 Start. Complete solution and verification follow on a separate page.

Question 1

Source: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 1

Primary concept: Ionic charge notation

Supporting concept(s): Formula anatomy

Builds on: Separate ionic charge from subscripts; Read a chemical formula before calculating

H0 — Attempt first

Attempt the original question independently before opening any hint or solution.

In the ion SO_4^{2-} , what does the superscript $2-$ represent?

- A. The number of oxygen atoms
- B. The overall ionic charge
- C. A coefficient for the species
- D. The number of sulfur atoms

Workspace

- target notation = _____
- subscript(s) = _____
- superscript charge = _____
- meaning = _____

H1 — Notice

Notice where the numeral is written: upper right or lower right?

H2 — Rule / model / representation

A superscript at the upper right gives the overall ionic charge; a subscript counts atoms within the formula.

H3 — Start

Circle the superscript $2-$ and label it 'species charge' before evaluating the options.

Reasoning route

- Read the given representation exactly as written or shown.
- Identify the chemical entities that must be tracked.
- Read coefficients, subscripts, charges and sides in their correct positions.
- Classify only after the decisive evidence is established.
- Check that the final conclusion follows from the evidence and rule used.

Transfer item 1 of 5 — source-faithful, attempted before any support is opened.

Anatomy of the formula

SUBSCRIPT 4

atoms inside one species



IONIC CHARGE $2-$

net charge on the whole ion

Notation position and its local chemical meaning.

Question 1 — Complete solution

- Read the ion exactly as written and separate the subscript from the superscript.
- Identify 4 as the oxygen-count subscript and 2^- as the overall ionic charge.
- Compare that meaning with every option before selecting the answer.

Verification

- Check that the same chemical species is being tracked.
- Check that the overall charge is preserved or interpreted correctly.

Final answer: B

The overall ionic charge

No additional condition or exception is recorded for this source item.

Question 2

Source: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 1

Primary concept: Particle-to-symbol translation

Supporting concept(s): none

Builds on: Translate between particles and symbols; Read a chemical formula before calculating

H0 — Attempt first

Attempt the original question independently before opening any hint or solution.

A particle diagram shows two separate H₂O particles. Which symbolic representation matches the diagram?

- A. H₂O
- B. 2H₂O
- C. H₄O₂
- D. 2H₂ + O₂

Source figure — particle model

School-level particle-count model — composition and count only.



● H × 4 ● O × 2

2 particle(s) shown

Workspace

- particle identity = _____
- number of separate particles = _____
- symbol for one particle = _____
- combined symbolic form = _____

H1 — Notice

Count the separate H₂O particles in the diagram before choosing a symbolic form.

H2 — Rule / model / representation

A leading coefficient counts whole particles or formula units; changing a subscript changes the composition of one particle.

H3 — Start

Write H₂O once for each drawn particle, then combine identical particles with one leading coefficient.

Reasoning route

- Identify whether the information is macroscopic, particulate or symbolic.
- Identify the chemical entities that must be tracked.
- Translate one representation into another without changing identity or count.
- State what the representation means chemically.
- Check that the final conclusion follows from the evidence and rule used.

Transfer item 2 of 5 — source-faithful, attempted before any support is opened.

Particle view of the same substance

School-level particle-count model — composition and count only.



● H × 4 ● O × 2

2 particle(s) shown

Question 2 — Complete solution

- Identify each drawn particle as H_2O without changing its internal composition.
- Count two separate H_2O particles.
- Represent the count with a leading coefficient, giving $2\text{H}_2\text{O}$ rather than changing subscripts.

Verification

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Final answer: B

$2\text{H}_2\text{O}$

No additional condition or exception is recorded for this source item.

Question 3

Source: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 1

Primary concept: Atom conservation

Supporting concept(s): none

Builds on: Check atom conservation explicitly; Read a chemical formula before calculating

H0 — Attempt first

Attempt the original question independently before opening any hint or solution.

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, which statement correctly verifies atom conservation?

- A. There are 2 H and 1 O atoms on each side.
- B. There are 4 H and 2 O atoms on each side.
- C. Only hydrogen is conserved.
- D. The coefficients do not affect atom counts.

Workspace

- left-side atom counts = _____
- right-side atom counts = _____
- first mismatch, if any = _____
- conservation conclusion = _____

H1 — Notice

Count H and O atoms separately on both sides of the equation.

H2 — Rule / model / representation

A coefficient multiplies every atom in the formula that follows it.

H3 — Start

Make a two-row ledger for H and O and calculate the left-side and right-side counts before reading the options.

Reasoning route

- Read coefficients, subscripts, charges and sides in their correct positions.
- Identify the chemical entities that must be tracked.
- Apply the required conservation check.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Transfer item 3 of 5 — source-faithful, attempted before any support is opened.

Atom ledger before and after



Element	Before	After	Matched?
H	4 	4 	✓ same
O	2 	2 	✓ same

Fill or compare the required ledger and locate the first mismatch.

Question 3 — Complete solution

- Apply each coefficient to the complete formula that follows it.
- Count 4 H and 2 O atoms on the reactant side.
- Count 4 H and 2 O atoms on the product side and compare the ledgers.

Verification

- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Final answer: B

There are 4 H and 2 O atoms on each side.

No additional condition or exception is recorded for this source item.

Question 4

Source: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 2

Primary concept: Reaction conditions

Supporting concept(s): none

Builds on: Keep reaction conditions attached

H0 — Attempt first

Attempt the original question independently before opening any hint or solution.

The source represents $A \rightarrow B$ with heat written over the reaction arrow. Which restatement preserves the source condition?

- A. A forms B with the recorded heat condition retained.
- B. A forms B and the heat condition may be omitted.
- C. Heat is a product of the reaction.
- D. Heat is a coefficient multiplying A.

Recorded condition: heat

Workspace

- reaction = _____
- recorded condition = _____
- where the condition belongs = _____
- condition-preserving statement = _____

H1 — Notice

Inspect the annotation attached to the reaction arrow; it is part of the source meaning.

H2 — Rule / model / representation

A condition written over or with a reaction arrow is neither a reactant nor a product and must stay attached to the reaction description.

H3 — Start

Copy the reaction meaning while explicitly retaining the word 'heat' as the recorded condition.

Reasoning route

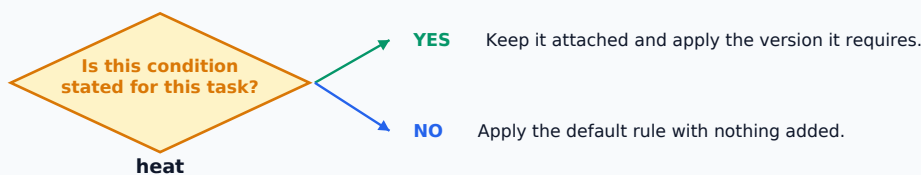
- Read the given representation exactly as written or shown.
- Read coefficients, subscripts, charges and sides in their correct positions.
- Check condition or exception.
- State what the representation means chemically.
- Check that every stated condition or exception is still attached.

Transfer item 4 of 5 — source-faithful, attempted before any support is opened.

Condition gate before the rule applies



The recorded condition stays attached to the arrow.



Question 4 — Complete solution

- Read the reaction and the arrow annotation as one source representation.
- Identify heat as a recorded reaction condition, not as a reactant, product or coefficient.
- Choose the statement that retains the heat condition with the reaction.

Verification

- Check that every stated condition or exception is still attached.

Final answer: A

A forms B with the recorded heat condition retained.

Preserve and verify the recorded condition: heat.

Question 5

Source: Synthetic External Corpus | 2026 | session SYNTHETIC | shift 2

Primary concept: Reacting-species roles

Supporting concept(s): none

Builds on: Attach a role only after tracking the species; Track the same reacting species

H0 — Attempt first

Attempt the original question independently before opening any hint or solution.

For $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, the half-change $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is shown. Which reactant is the reducing agent?

- A. Zn
- B. Cu^{2+}
- C. Zn^{2+}
- D. Cu

Workspace

- species A: _____ $\rightarrow$ _____
- species B: _____ $\rightarrow$ _____
- electron-losing species = _____
- electron-gaining species = _____
- reducing agent = _____
- oxidising agent = _____

H1 — Notice

Use the shown half-change and look at which side contains electrons.

H2 — Rule / model / representation

A reactant that loses electrons is oxidised and acts as the reducing agent because it supplies electrons to the other species.

H3 — Start

From $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$, mark Zn as the electron-losing reactant before assigning the agent name.

Reasoning route

- Identify the chemical entities that must be tracked.
- Read coefficients, subscripts, charges and sides in their correct positions.
- Track the same species or state from before to after.
- Classify only after the decisive evidence is established.
- State what the representation means chemically.
- Check that the final conclusion follows from the evidence and rule used.

Transfer item 5 of 5 — source-faithful, attempted before any support is opened.

Sort the species before naming any role



Reactant species	Zn Cu^{2+}
Species that changes	fill only from the evidence above
Spectator (if present)	fill only from the evidence above
Product species	Zn^{2+} Cu

Attach a role only after identifying the changing reactant species.

Question 5 — Complete solution

- Track the reactant species through the shown half-change.
- Zn appears on the reactant side and produces electrons, so Zn loses electrons and is oxidised.
- The electron-losing reactant is the reducing agent; attach the role to Zn only after that change is established.

Verification

- Check that the same chemical species is being tracked.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the final conclusion follows from the evidence and rule used.

Final answer: A

Zn

No additional condition or exception is recorded for this source item.